

Cover Letter — *Fractal and Fractional*

Dear Editors,

I am pleased to submit our manuscript, “**Spectral Isomorphism between Renormalization Flow in Non-Autonomous Quadratic Maps and the Riemann Zeros: A Numerical Study,**” for consideration as a research article in *Fractal and Fractional*.

Significance and context. Finding a dynamical system whose spectrum reproduces the non-trivial Riemann zeros is a long-standing question in mathematical physics (the Hilbert–Pólya conjecture). Existing analytical approaches, from Berry–Keating’s $H = xp$ Hamiltonian to random-matrix models, elegantly capture the macroscopic statistics of high-order zeros but rarely generate the discrete low-order zeros deterministically. We take a different, “bottom-up” route: building on our recently published result that the symbolic dynamics of the Logistic map at its band-merging point is isomorphic to the prime sieve, we introduce a non-autonomous quadratic map governed by a logarithmic renormalization-flow schedule and study its transfer-operator spectrum. The model reproduces the low-order Riemann zeros with small numerical error and exhibits a change of behavior near $N \approx 20$ that shows a qualitative spatial coincidence with the enlarged error bars reported in recent USTC ion-trap quantum simulations of the same zeros. At large N , an ablation study shows that restoring the conjugate spectrum drives the fitted ground-state intercept to zero, pointing to single-sided truncation as a likely source of the residual envelope.

Why *Fractal and Fractional*. The manuscript centers on renormalization-flow scaling, non-autonomous (time-dependent) discrete dynamics, and the fractal/self-similar structure of the logistic map’s band-merging cascade, connecting nonlinear dynamical-systems theory to a classical problem in analytic number theory. This combination of fractal dynamics, scaling laws, and numerical spectral analysis fits well within the journal’s scope.

On rigor and claims. We are deliberately explicit about epistemic status throughout the manuscript: results are presented as numerical observations and heuristic arguments, not proofs, and a dedicated table classifies every main claim as established, published, numerical, heuristic, qualitative, or conjectural. We do not claim to have identified the Hilbert–Pólya operator or to have proven any statement about the Riemann Hypothesis. All code and data are openly available at https://github.com/maris205/riemann_logistic for full reproducibility.

Prior submissions. This manuscript has not been previously submitted to any MDPI journal.

We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with or published in another journal.

All authors have approved the manuscript and agree with its submission to *Fractal Fract.*

Thank you for considering our submission. We would be happy to suggest suitable referees with expertise in nonlinear dynamics, ergodic theory, and analytic number theory upon request.

Sincerely,

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